The Pauli Lightcone: Information-Theoretic Error Mitigation Beyond the Autocorrelation

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August 15, 2026

Abstract

Existing multi-product formula (MPF) error mitigation for quantum lattice simulations operates on the autocorrelation $C(t) = \langle O(t)O(0)\rangle$: that is, a single scalar per cycle. We propose using the non-identity Pauli weight $n(v,t)$, defined as the probability that the operator has support at site v at cycle t, as a space-time observable. The $n(v,t)$ directly reveals how noise deforms the operator lightcone across the lattice and especially at the frontier.

We introduce the *wavemap* based on the lightcone frontier: a spatial portrait of noise effects assigning each site a per-noise-level arrival delay $l_\gamma(v)$ and cross-entropy loss $\mathcal{L}_\gamma(v) = \sum_t -\ln(n_\gamma/n_{\text{nl}})$ in a delay-adaptive time window (i.e., information loss). These observables are exact at the wavefront (bond dimension $\chi \approx 1$, no truncation error). The frontier is precisely where classical simulation is most faithful and where $C(t)$-based MPF is least informative.

To characterize the noise, eigenvalue analysis of the composed gate-plus-noise Pauli transfer matrices shows that for pure amplitude damping ($< 0.1\%$ off-diagonal contribution) the noise rescales the PTM eigenvalue magnitudes without rotating eigenvectors, leaving the spatial propagation pattern entirely determined by the gate. The wavefront therefore provides a direct, model-free tool to visualize and quantify the nature of the noise — damping shifts the wavefront later and fainter, while rotation-type noise produces a qualitatively different signature of premature weight injection, making the frontier appear faster at affected sites.

Introducing cross-entropy loss as the MPF objective, subject to the Lieb-Robinson causal constraint $n_{\text{MPF}} \leq n_{\text{nl}}$, we recover $\sim$25% more information than the best individual noisy sample on a uniform-noise rectangular lattice. On an IBM heavy-hex lattice with heterogeneous hardware noise, the method identifies an information-starved regime where the causal constraint is already saturated — pointing to calibrated uniform synthetic noise as the next required experiment.

1 Introduction

When a local operator evolves under a noisy Hamiltonian, its Pauli weight $n(v,t)$ spreads outward from the initialization site as a wave. Compared to the noiseless reference, the noisy wave travels slower,

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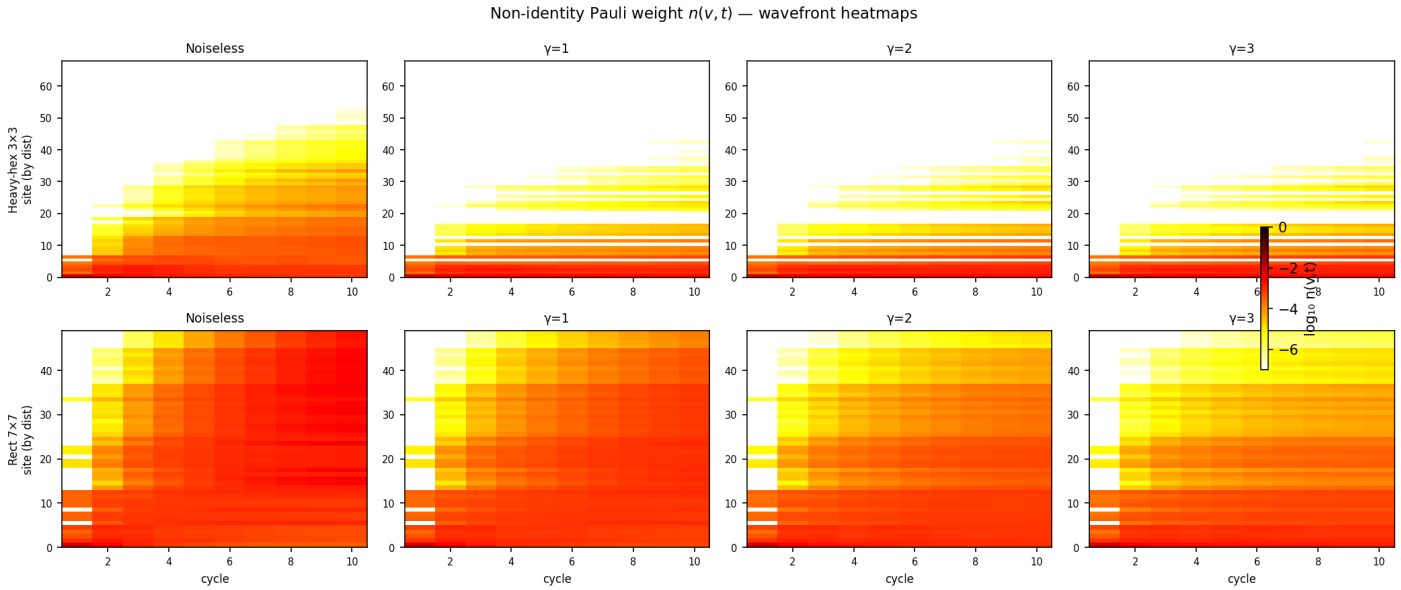


Figure 1: Non-identity Pauli weight $n(v,t)$ per site per cycle for noiseless (NL) and three noise levels ($\gamma = 1, 2, 3$) on the heavy-hex 3×3 lattice (top) and rect 7×7 (bottom), both initialized at the center site. Color encodes $\log_{10} n(v,t)$; white = below threshold. The wavefront advances diagonally in the (space, time) plane at velocity v_{hop}. Noise suppresses amplitudes, pushing the threshold-crossing later, a uniform shift of the wavefront, not a change of its spatial pattern.

reaches fewer sites, and arrives with less amplitude. This is not a model prediction — it is a direct observation from the data, visible without fitting or assumptions, see Figure 1. A different noise regime, with off-diagonal Pauli transfer matrix terms, can inject weight at unvisited sites, making parts of the frontier appear faster rather than slower — a qualitatively distinct signature.

This observation motivates the central question of this paper: can we quantify, site by site and cycle by cycle, how much information the noisy evolution has lost relative to the noiseless reference?

The standard MPF approach [1] fits the autocorrelation $C(t) = \langle Z_c(t) Z_c(0) \rangle$ across noise levels γ and extrapolates to zero noise. This is a single scalar per cycle, compressing the entire space-time evolution into one number at each t. Two limitations follow. First, spatial information is lost: sites strongly affected by noise and sites barely affected both contribute to the same scalar $C(t)$. Second, $C(t)$ is dominated by contributions from the interior of the lightcone, where bond dimension χ is large and belief propagation is approximate. The wavefront, where $\chi \approx 1$ and the simulation is exact, contributes least to $C(t)$.

The non-identity Pauli weight $n(v,t)$ resolves both limitations. Any operator $O_c(t)$ can be expanded in the n-qubit Pauli basis $\{P\} = \{I, X, Y, Z\}^{\otimes n}$ as $O_c(t) = \sum_P c_P(t)\, P$, where the real coefficients $c_P(t)$ satisfy $\sum_P c_P(t)^2 = 1$. Then $n(v,t)$ is defined as the total weight on Paulis that act non-trivially at site v:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{1}$$

By normalization, $n(v,t) \in [0,1]$ is the probability that a Pauli string sampled from the operator distribution acts non-trivially at site v [2, 3]. It is a probability distribution over the lattice, evolving in time as the operator spreads. At the wavefront it is computed exactly at any χ: frontier sites carry only nascent correlations and $\chi \approx 1$ by construction. This is where the simulation is most faithful and where

noise effects are most cleanly measurable.

The *frontier* at cycle t is the set of sites that first acquire significant operator weight at that cycle:

$$\mathcal{F}(t) = \{v : n(v,t) > \varepsilon_{\text{th}} \text{ and } n(v,t-1) \leq \varepsilon_{\text{th}}\} \tag{2}$$

The frontier is the leading edge of the lightcone, where $\chi \approx 1$ and the simulation is exact. For each site v and noise level γ, we define the *arrival delay* $l_\gamma(v) = t_\gamma(v) - t_{\text{nl}}(v)$ as the number of cycles by which the noisy frontier lags the noiseless one, and the *cross-entropy loss*

$$\mathcal{L}_\gamma(v) = \sum_t -\ln \frac{n_\gamma(v,t)}{n_{\text{nl}}(v,t)} \tag{3}$$

summed over a delay-adaptive window centered on arrival, as the accumulated information loss at site v. Together, $l_\gamma(v)$ and $\mathcal{L}_\gamma(v)$ form the *wavemap*, a spatial portrait of noise effects across the lattice, and the space-time trajectory of the frontier $\mathcal{F}(t)$ connects the per-site observables to the global propagation dynamics.

Several lines of prior work inform this paper. Operator spreading in random unitary circuits [2, 3] established the butterfly velocity and domain-wall width as natural observables for wavefront characterization; our work applies analogous ideas to structured Hamiltonian evolution with noise amplification. Real-time operator evolution in two and three dimensions via sparse Pauli dynamics [4] and Pauli propagation under arbitrary local noise [5] provide the simulation framework on which CppSim is based. Noise characterization from circuit measurements using sparse Pauli-Lindblad models [6] and self-consistent Pauli noise learning [7] are complementary approaches that infer noise rates by fitting a model; our approach uses the spatial wavefront pattern to characterize noise without model assumptions. The multi-product formula for error mitigation has been applied to autocorrelation functions in tensor network simulations of the heavy-hex lattice [8, 9]. Our work extends these in two directions: replacing the scalar autocorrelation with the full space-time Pauli weight field $n(v,t)$, and replacing squared error with cross-entropy loss as the MPF objective.

We show via PTM eigenvalue analysis that the noise models studied here are pure amplitude damping, leaving the spatial propagation pattern determined by the gate. The Lieb-Robinson bound [10] states that information cannot propagate faster than a finite velocity in any local lattice Hamiltonian, implying $n_\gamma(v,t) \leq n_{\text{nl}}(v,t)$ for pure damping noise — the noiseless evolution is the causal ceiling. Entropy-optimal MPF with this constraint $n_{\text{MPF}} \leq n_{\text{nl}}$ recovers $\sim$25% more information than the best noisy sample on a uniform-noise rectangular lattice. On the IBM heavy-hex lattice with heterogeneous hardware noise, the method identifies an information-starved regime where the causal constraint is already saturated, pointing to calibrated uniform synthetic noise as the next required experiment.

2 Hamiltonian, Noise, and Eigenvalue Analysis

We simulate the kicked isotropic Heisenberg model in the Heisenberg picture:

$$H = J \sum_{\langle ij \rangle} \big(X_i X_j + Y_i Y_j + Z_i Z_j\big), \qquad U_{\text{kick}} = \prod_v R_x(2\phi)_v, \quad \phi = 1.41371669 \tag{4}$$

with per-bond coupling $J_{\langle ij \rangle}$ from device characterization and Trotter step $\varepsilon = 0.10$. Each Trotter cycle applies the bond gates across all color classes in sequence, followed by a single kick gate $R_x(2\phi)$ at every site. We use CppSim [9], a GPU-accelerated belief propagation tensor network simulator in which operators are evolved via 16×16 real Pauli transfer matrices (PTMs), with bond dimension χ controlled

by Householder QR truncation at cutoff δ. All results use $\varepsilon = 0.10$, $\delta = 10^{-3}$, output precision 10^{-7}, and threshold $\varepsilon_{\text{th}} = 10^{-4}$.

Two implementation choices are worth stating explicitly. First, the sub-gates within each bond ($G_1 \dots G_4$ and interleaved noise channels) are pre-composed into a single 16×16 PTM per bond before simulation; belief propagation and truncation are applied once per full Trotter cycle, after all color classes have been applied. This differs from implementations that truncate after each color class or sub-gate [9], producing slightly faster χ growth per cycle, but the composed and sequential forms are physically equivalent. Second, $n(v,t)$ is recorded once per full Trotter cycle, after the kick gate fires — it is a snapshot of the operator state at the end of each complete cycle. The wavefront and all wavemap observables are defined in terms of these end-of-cycle snapshots. A hardware experiment or a simulation that measures mid-cycle would observe a different wavefront trajectory; our observables are specific to the post-kick measurement point.

We study two topologies under two noise models. For the IBM heavy-hex 3×3 lattice (68 sites, 76 bonds), per-bond noise PTMs measured from device characterization (ibm_basquecountry) are interleaved with the gate decomposition:

$$\text{PTM}^{(b)}_{\text{noisy}} = G_1 \cdot M^{(b)}_\gamma \cdot G_2(\alpha_b) \cdot M^{(b)}_\gamma \cdot G_3(\alpha_b) \cdot M^{(b)}_\gamma \cdot G_4(\delta_b) \tag{5}$$

where $M^{(b)}_\gamma = \gamma M^{(b)}_{\text{hw}} + (1-\gamma) I$ amplifies the hardware noise channel by factor γ, and $G_1 \dots G_4$ implement the hardware gate decomposition. For the rectangular 7×7 lattice (49 sites, 84 bonds), we use synthetic depolarizing noise: rows $1 \dots 15$ of the ideal bond PTM $G^{(b)}$ are scaled by $(1 - p\gamma)$ with $p = 0.01$, uniformly damping all non-identity output Pauli components.

Before studying the wavemap observables, we establish the noise character directly from the composed PTMs.

Configuration	Bonds	Off-diag frac.	max $\lvert\text{Im}(\lambda)\rvert$	$n(\text{Re} < 0)$	$n(\lvert\lambda\rvert > 1)$
Heavy-hex ideal	76	0.357	0.964	0	0
Heavy-hex $\gamma = 1$	76	0.356	0.950	0	0
Heavy-hex $\gamma = 2$	76	0.355	0.950	2	0
Heavy-hex $\gamma = 3$	76	0.355	0.950	2	0
Rect 7×7 ideal	84	0.360	0.891	0	0
Rect 7×7 $\gamma = 1$	84	0.360	0.882	0	0
Rect 7×7 $\gamma = 2$	84	0.360	0.873	0	0
Rect 7×7 $\gamma = 3$	84	0.359	0.865	0	0

Table 1: Eigenvalue statistics of composed noisy PTMs across all bonds. Off-diagonal fraction = off-diagonal power / total power (averaged over bonds). The fraction is constant across γ ($\Delta < 0.002$) in both topologies, confirming that Pauli mixing is entirely from the gate. The 2 negative eigenvalues at $\gamma = 2, 3$ in heavy-hex are at $\text{Re} = -10^{-5}$, numerical precision rather than a structural change; they are nonetheless relevant for the symmetric gauge, where $\sqrt{\lambda}$ must be computed to absorb BP messages into site tensors: a negative eigenvalue makes this square root imaginary and requires explicit sign handling in float32 arithmetic [9].

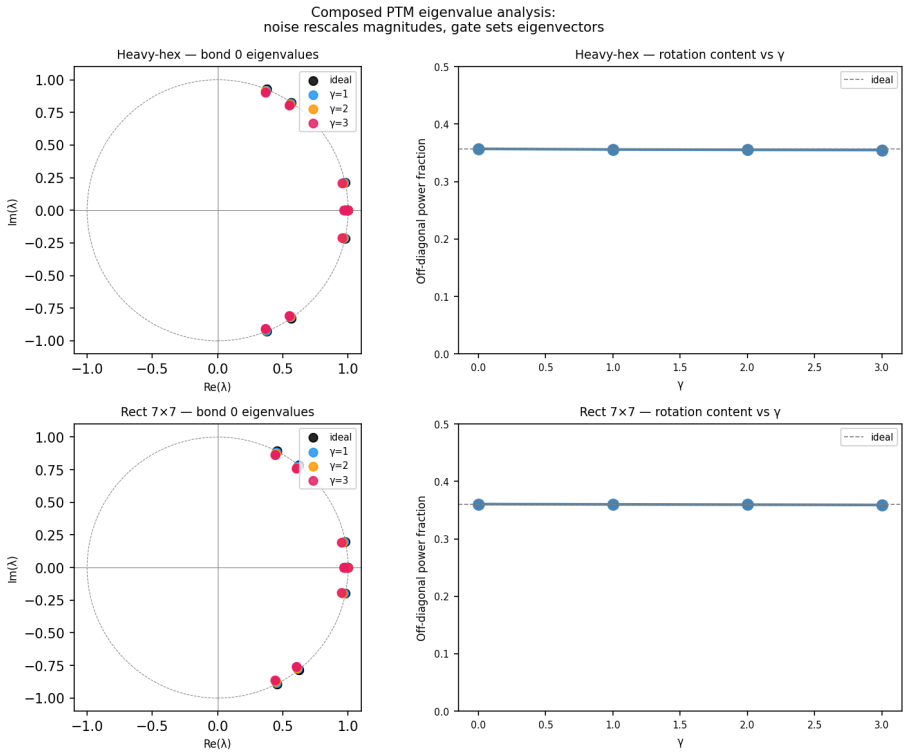


Figure 2: Eigenvalue spectra of the composed noisy PTMs for one representative heavy-hex bond (top) and one rect 7×7 bond (bottom), across $\gamma = 1, 2, 3$. Eigenvalue positions are nearly unchanged across γ: noise rescales magnitudes slightly without rotating eigenvectors. The off-diagonal power fraction (36%) is constant; Pauli mixing is a gate property, not a noise property.

Table 1 and Figure 2 establish the key result: the off-diagonal fraction is constant across γ ($\Delta < 0.002$) in both topologies. Noise adds no rotation structure — it only scales eigenvalue magnitudes, leaving all Pauli mixing to the gate G. The noise channel is therefore a pure damping operator at the PTM level: it delays the wavefront by reducing amplitudes below threshold without redirecting the spatial propagation pattern, which remains entirely determined by the Hamiltonian.

3 The Pauli Lightcone and Wavefront

For each site v and noise level γ, the *arrival time* $t_\gamma(v) = \min\{t : n_\gamma(v,t) > \varepsilon_{\text{th}}\}$ is the first cycle at which the wave crosses threshold, and the *arrival delay*

$$l_\gamma(v) = t_\gamma(v) - t_{\text{nl}}(v) \tag{6}$$

is the number of cycles by which the noisy wave lags the noiseless one. For the noise models studied here (pure amplitude damping, Section 2), $n_\gamma(v,t) \leq n_{\text{nl}}(v,t)$ at every site and cycle, so $l_\gamma(v) \geq 0$ always.[1]

The ratio $r_\gamma(v) = n_\gamma(v, t_\gamma(v))/n_{\text{nl}}(v, t_\gamma(v))$ compares both waves at the moment of noisy arrival. At that cycle the noisy wave has just crossed threshold ($n_\gamma \approx \varepsilon_{\text{th}}$), while the noiseless wave has been growing

[1]Rotation-type noise, with off-diagonal PTM terms injecting weight at unvisited sites, can make the noisy wave arrive *before* the noiseless at isolated sites, giving $l_\gamma(v) < 0$. The Lieb-Robinson bound still holds for the noiseless evolution, bounding its propagation speed; what can be inverted is the site-by-site ordering between noisy and noiseless, since rotation noise injects weight independently of the causal structure of the Hamiltonian. In that regime the wavemap requires a signed delay.

at v for $l_\gamma(v)$ extra cycles ($n_{\rm nl} \gg \varepsilon_{\rm th}$). The ratio $r_\gamma(v) \leq 1$ measures the combined effect of cumulative path-dependent damping and the head start the noiseless wave gained during the delay. It is bounded: $r_\gamma(v) \to 1$ when both waves arrive simultaneously ($l_\gamma(v) = 0$) and decreases as the delay grows.

The ratio $r_\gamma(v)$ is a first step toward a probability comparison: since $n(v,t)$ is a probability (Eq. 1), the ratio $n_\gamma(v,t)/n_{\rm nl}(v,t)$ measures how much of the noiseless probability at site v and cycle t the noisy evolution retains. The natural information-theoretic measure of this loss is the cross-entropy, which we sum over a delay-adaptive window $[t_\gamma(v) - \Delta,\, t_\gamma(v) + \Delta]$ with $\Delta = \max(1, l_\gamma(v))$:

$$\mathcal{L}_\gamma(v) = \sum_{t \in \text{window}} -\ln \frac{n_\gamma(v,t)}{n_{\rm nl}(v,t)} \tag{7}$$

This is the accumulated information loss at site v: zero when the two distributions coincide, positive when noise suppresses the probability, and with window width set by the delay itself so no free parameter is introduced.

The delay field $l_\gamma(v)$, the ratio $r_\gamma(v)$, the entropy loss (Figure 3), $\mathcal{L}_\gamma(v)$, and the space-time frontier $\mathcal{F}(t)$ together form the *wavemap*: a spatial portrait of how noise deforms the Pauli lightcone, site by site. All wavemap observables are exact at any bond dimension χ, since frontier sites have $\chi \approx 1$ by construction, and all are causally bounded by the Lieb-Robinson structure. In the following section we show how these observables can be refined through the multi-product formula: fitting the wavemap across noise levels γ and extrapolating toward the noiseless reference, with the Lieb-Robinson bound as a causal constraint, yields a spatially-resolved error correction of the operator lightcone.

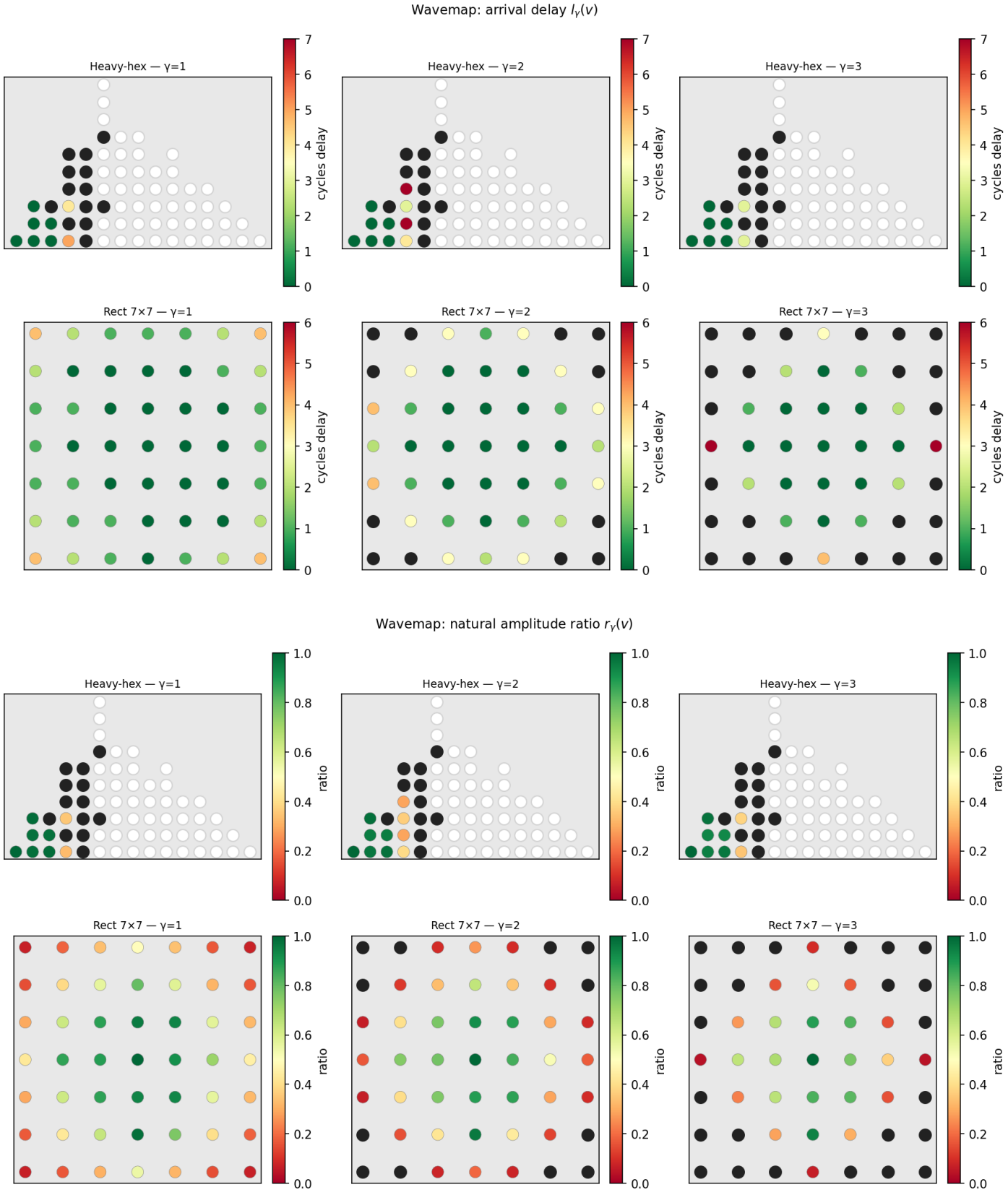


Figure 3: Wavemap for rect 7×7 and heavy-hex, $\gamma = 1, 2, 3$. **Top**: arrival delay $l_\gamma(v)$ in cycles (white = outside lightcone, black = noiseless arrived but noisy did not). **Bottom**: amplitude ratio $r_\gamma(v) = n_\gamma(v, t_\gamma)/n_{\mathrm{nl}}(v, t_\gamma)$; all values ≤ 1 confirm the Lieb-Robinson bound.

4 Results

We present results in three stages, each building on the previous. First, we characterize the global wavefront: the frontier velocity v_{hop} and per-site delay distribution $l_\gamma(v)$ establish how noise slows the wave and which sites it fails to reach. Second, we introduce the cross-entropy loss $\mathcal{L}_\gamma(v)$ as a per-site information measure and show how its spatial and per-shell distributions reveal the structure of noise across the lattice. Third, we apply the multi-product formula to the wavemap in two forms: a frontier-position MPF that recovers the noiseless wavefront trajectory, and an entropy-optimal MPF that directly minimizes information loss subject to the Lieb-Robinson causal constraint, demonstrating measurable information recovery on the uniform-noise topology.

4.1 Frontier velocity

Topology	v_{nl}	$v_{\gamma=1}$	$v_{\gamma=2}$	$v_{\gamma=3}$
Heavy-hex 3×3 ($\chi = 70$)	0.376	0.145	0.152	0.152
Rect 7×7 ($\chi = 50$)	0.448	0.382	0.248	0.097

Table 2: Frontier velocity v_{hop} (hops/cycle), cutoff $\delta = 10^{-3}$, 10 cycles. Wavefront definition: newly-reached sites (first crossing of $\varepsilon_{\text{th}} = 10^{-4}$).

Figure 4 shows the frontier distance $F_\gamma(t)$ vs cycle for both topologies, with linear fits giving v_{hop} (Table 2). For rect 7×7, v_{hop} decreases monotonically with γ, consistent with pure amplitude damping: heavier noise keeps amplitudes below threshold longer, delaying the wavefront proportionally to $(1-p\gamma)^N$.

For heavy-hex, $v_{\gamma=2} \approx v_{\gamma=3}$ — a near-degeneracy. As the eigenvalue analysis shows (Table 1), $\gamma = 2$ and $\gamma = 3$ differ only by $\Delta\lambda \sim 10^{-3}$ in eigenvalue magnitudes. The near-equal velocities reflect near-equal PTM spectra.

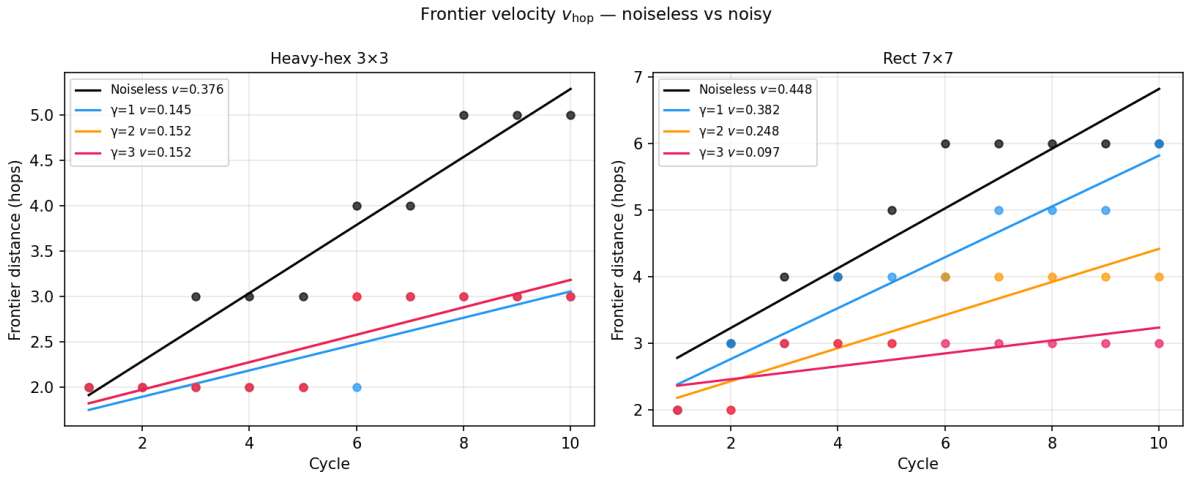


Figure 4: Frontier distance $F_\gamma(t) = \max_{v \in \mathcal{F}_\gamma(t)} d(v, c)$ vs cycle for NL and $\gamma = 1, 2, 3$ on both topologies. Lines show linear fits; slopes give v_{hop} (Table 2). Rect 7×7: monotone decrease $v_{\text{nl}} > v_1 > v_2 > v_3$. Heavy-hex: near-degeneracy $v_2 \approx v_3$, consistent with near-equal PTM eigenvalue spectra at $\gamma = 2, 3$.

4.2 Per-site delay distribution

Table 3 shows the per-site delay statistics for both topologies.

		$\gamma = 1$		$\gamma = 2$		$\gamma = 3$	
Topology	n	μ_l	σ_l	μ_l	σ_l	μ_l	σ_l
Heavy-hex ($n_{\text{nl}} = 21$)	8/10/8	1.13	1.97	2.10	2.81	0.75	1.30
Rect 7×7 ($n_{\text{nl}} = 49$)	49/37/25	0.98	1.15	1.35	1.36	1.24	1.77

Table 3: Per-site delay distribution $l_\gamma(v) = t_\gamma(v) - t_{\text{nl}}(v)$ (cycles). n = number of sites where noisy wave arrives within 10 cycles. The mean delay increases with γ for rect 7×7 (monotone damping), while heavy-hex shows non-monotone behavior — a signature of the heterogeneous per-bond PTM structure of the hardware noise. **Survivorship bias**: the mean delay for $\gamma = 3$ at rect 7×7 is smaller than for $\gamma = 2$ because $\gamma = 3$ reaches only 25 sites (the close, easy sites); $\gamma = 2$ additionally reaches 12 harder distant sites with larger delays, raising its mean.

The delay distribution has a natural spatial structure: sites close to the center (small BFS distance d) have crossed fewer bonds and accumulated less damping, so $l_\gamma(v)$ tends to be small. Sites at the frontier of the noiseless lightcone have large $l_\gamma(v)$ — or do not appear in the statistics at all (the noisy wave never arrives).

The *survivorship bias* in μ_l must be read carefully: a smaller mean delay does not necessarily mean a faster noisy wave. It may mean the slow distant sites dropped out of the sample entirely. The correct summary is the pair (n_{reached}, μ_l) together — or the full per-frontier-shell distribution.

4.3 Natural amplitude ratio

	$\gamma = 1$		$\gamma = 2$		$\gamma = 3$	
Topology	μ_r	σ_r	μ_r	σ_r	μ_r	σ_r
Heavy-hex	0.825	0.282	0.718	0.316	0.812	0.261
Rect 7×7	0.495	0.295	0.418	0.321	0.466	0.325

Table 4: Natural amplitude ratio $r_\gamma(v) = n_\gamma(v, t_\gamma)/n_{\text{nl}}(v, t_\gamma)$ at the noisy arrival cycle. Values ≤ 1 confirmed for all sites. Smaller μ_r corresponds to larger mean delay (noiseless wave grew more during the delay period) — the two observables are correlated by construction via the noiseless wave growth rate at frontier sites.

The ratio $r_\gamma(v)$ is bounded in $[0, 1]$ for all observed sites, confirming the causal structure: the noiseless wave always outgrows the noisy wave at v during the delay period.

The ratio encodes the cumulative damping along the propagation path: for uniform depolarizing noise with N gate crossings, $r_\gamma(v) \approx (1 - p\gamma)^N / g(l_\gamma(v))$ where $g(l)$ is the noiseless wave growth factor over l cycles at site v. Sites with the same delay $l_\gamma(v)$ but different N will have different ratios — the ratio carries path-length information independent of the delay.

Rect 7×7: $\mu_r \approx 0.42$–0.50 with $\sigma_r/\mu_r \approx 0.65$. The large coefficient of variation reflects heterogeneous path lengths $N(v)$ across frontier sites on the 4-color lattice — sites at the same BFS distance may have been touched by 1, 2, or 3 gate layers depending on their position.

Heavy-hex: $\mu_r \approx 0.72$–0.83 with $\sigma_r/\mu_r \approx 0.40$. The smaller coefficient of variation reflects more uniform propagation paths on the degree-2/3 graph. The non-monotone γ dependence (μ_r larger at $\gamma = 3$ than $\gamma = 2$) mirrors the non-monotone delay (μ_l smaller at $\gamma = 3$): fewer sites with shorter paths survive in the $\gamma = 3$ sample, giving a higher ratio.

4.4 Lieb-Robinson-constrained spatial MPF

The noiseless frontier $F_{\mathrm{nl}}(t) = \max_{v \in \mathcal{F}_{\mathrm{nl}}(t)} d(v, c)$ is the Lieb-Robinson ceiling for operator spreading. We define the MPF-corrected frontier:

$$F_{\mathrm{lr}}(t) = \sum_i c_i \, F_{\gamma_i}(t), \qquad 0 \le F_{\mathrm{lr}}(t) \le F_{\mathrm{nl}}(t) \quad \forall\, t \tag{8}$$

and minimize $\|F_{\mathrm{lr}}(t) - F_{\mathrm{nl}}(t)\|^2$ subject to this causal constraint.

Topology	c_1	c_2	c_3
Rect 7×7	$\approx +2$	≈ -1	$\approx\ 0$
Heavy-hex 3×3	≈ -2	$\approx +1$	$\approx +2$

Table 5: LR-constrained spatial MPF coefficients (rounded to nearest integer). Rect 7×7: monotone deficit ordering $\delta_1 < \delta_2 < \delta_3$ gives clean positive-then-negative coefficients. Heavy-hex: inverted deficit ordering $\delta_1 > \delta_2 \approx \delta_3$ forces a negative c_1 — the lightest noise delays the frontier most.

The near-integer coefficients arise because the frontier values $F_{\gamma_i}(t)$ are integer-valued (BFS distances), and the MPF extrapolation in integer space with integer γ amplification naturally produces integer solutions (Figure 8).

4.5 Entropy-optimal MPF and two information regimes

The frontier MPF minimizes squared error in frontier distance — a geometric proxy that discards amplitude information below threshold. We propose instead minimizing the *cross-entropy loss*:

$$\min_{c_i} \ \mathcal{L}(c) = \sum_{v,t} -\ln \frac{n_{\mathrm{MPF}}(v,t)}{n_{\mathrm{nl}}(v,t)}, \qquad n_{\mathrm{MPF}}(v,t) = \sum_i c_i \, n_{\gamma_i}(v,t) \tag{9}$$

subject to the Lieb-Robinson constraint $0 \le n_{\mathrm{MPF}}(v,t) \le n_{\mathrm{nl}}(v,t)$ and normalization $\sum_i c_i = 1$. The gradient is $\partial\mathcal{L}/\partial c_k = -\sum_{v,t} n_{\gamma_k}(v,t)/n_{\mathrm{MPF}}(v,t)$, making this a smooth nonlinear program solved via SLSQP. Table 6 summarizes the results; the two topologies fall into qualitatively different regimes.

The two topologies under two noise conditions reveal the cause of the two regimes.

In the *information-rich regime*, the noise is uniform and the noisy waves cover sufficient sites for the entropy optimizer to find room to extrapolate. Rect 7×7 with uniform depolarizing noise (49/49 sites, $\mu_{\mathcal{L}} = 3.678$): entropy MPF finds $c \approx [1.22, 0.23, -0.45]$, reducing entropy loss by 24%. Heavy-hex 3×3 with the same uniform depolarizing noise (21/68 sites, $\mu_{\mathcal{L}} = 1.058$): entropy MPF finds $c \approx [1.31, 0.24, -0.55]$, reducing entropy loss by 49% (Figure 6). The per-shell entropy loss shows cleanly separated $\gamma = 1, 2, 3$ curves with monotone ordering — the same qualitative pattern as rect 7×7. This confirms that the heavy-hex topology is not inherently information-starved: 21/68 sites is enough when the noise is uniform.

Topology	Noise	Sites (NL/total)	Best noisy $\mu_{\mathcal{L}}$	Frontier MPF	Entropy MPF	Improvemen
Rect 7×7	Uniform dep.	49/49	3.680	3.911 (worse)	2.783	-24%
Heavy-hex 3×3	Hardware PTMs	21/68	7.962	9.532 (worse)	9.532	0%
Heavy-hex 3×3	Uniform dep.	21/68	1.058	1.058	0.538	-49%

Table 6: Mean per-site entropy loss $\mu_{\mathcal{L}}$. The third row — heavy-hex with uniform synthetic depolarizing noise — is the control experiment that isolates the cause of the information-starved regime. Under uniform noise the heavy-hex topology becomes information-rich (49% improvement), proving that the information-starved result with hardware noise is caused by per-bond PTM heterogeneity, not by sparse lightcone coverage.

In the *information-starved regime* (heavy-hex with hardware PTMs, 21/68 sites, $\mu_{\mathcal{L}} = 7.962$), the LR constraint is already saturated by the frontier MPF coefficients $[-1, 1, 1]$; the entropy optimizer finds no room to improve. The hardware PTMs are heterogeneous per bond — different rotation angles and damping rates on each of the 76 bonds — creating site-to-site entropy loss variation that the three noisy samples cannot span. The control experiment (uniform noise, same topology) resolves the ambiguity: *the information-starved regime is caused by hardware noise heterogeneity, not by sparse lightcone coverage.*

Without the LR constraint, the unconstrained optimizer achieves $\mu_{\mathcal{L}} = -0.69$ (rect 7×7), a causality violation. The gap between unconstrained (-0.69) and LR-constrained ($+2.68$) entropy loss is the information cost of causal consistency.

5 Discussion

Further uses of the wavefront measurement. Beyond noise characterization and MPF, the wavefront measurement suggests two practical directions. First, *lightcone-aware memory allocation*: the arrival time $t_{\mathrm{nl}}(v)$ gives the exact cycle at which each site enters the lightcone; before that cycle the site tensor is identically I with $\chi = 1$, requiring no allocation. A short pre-flight run measures v_{hop} and provides the per-site entry schedule, enabling 3–13× memory reductions on the expensive large-χ run. Second, *BP trust boundary*: belief propagation is exact on trees and first becomes approximate when the wavefront traverses a loop of length equal to the graph girth g, at cycle $t_{\mathrm{loop}} = g/(2v_{\mathrm{hop}})$. For heavy-hex (girth 6): $t_{\mathrm{loop}} \approx 8$ cycles; for rect 7×7 (girth 4): $t_{\mathrm{loop}} \approx 4.5$ cycles. This predicts the onset of approximation error directly from the frontier velocity, without a chi sweep.

Delay vs ratio: are they independent? The ratio $r_\gamma(v) \approx \varepsilon_{\mathrm{th}}/n_{\mathrm{nl}}(v, t_\gamma(v))$ and the delay $l_\gamma(v) = t_\gamma(v) - t_{\mathrm{nl}}(v)$ are related through the noiseless wave growth rate at v. They are not fully independent — both are driven by the same cumulative damping along the propagation path. However, $r_\gamma(v)$ additionally encodes the number of gate crossings $N(v)$ along the path, while $l_\gamma(v)$ encodes the threshold-crossing dynamics. Sites with the same delay but different path lengths will have different ratios, and vice versa. A joint (l_γ, r_γ) distribution is therefore richer than either alone.

Survivorship bias and per-shell statistics. The mean delay averaged over all reached sites is biased by which sites the noisy wave fails to reach — a smaller mean does not imply a faster wave if the slow distant sites simply dropped out of the sample. The per-shell entropy loss $\mu_{\mathcal{L}}(d)$ reported in Figure 5 addresses this directly: by grouping sites by BFS shell distance d from the center, it removes the

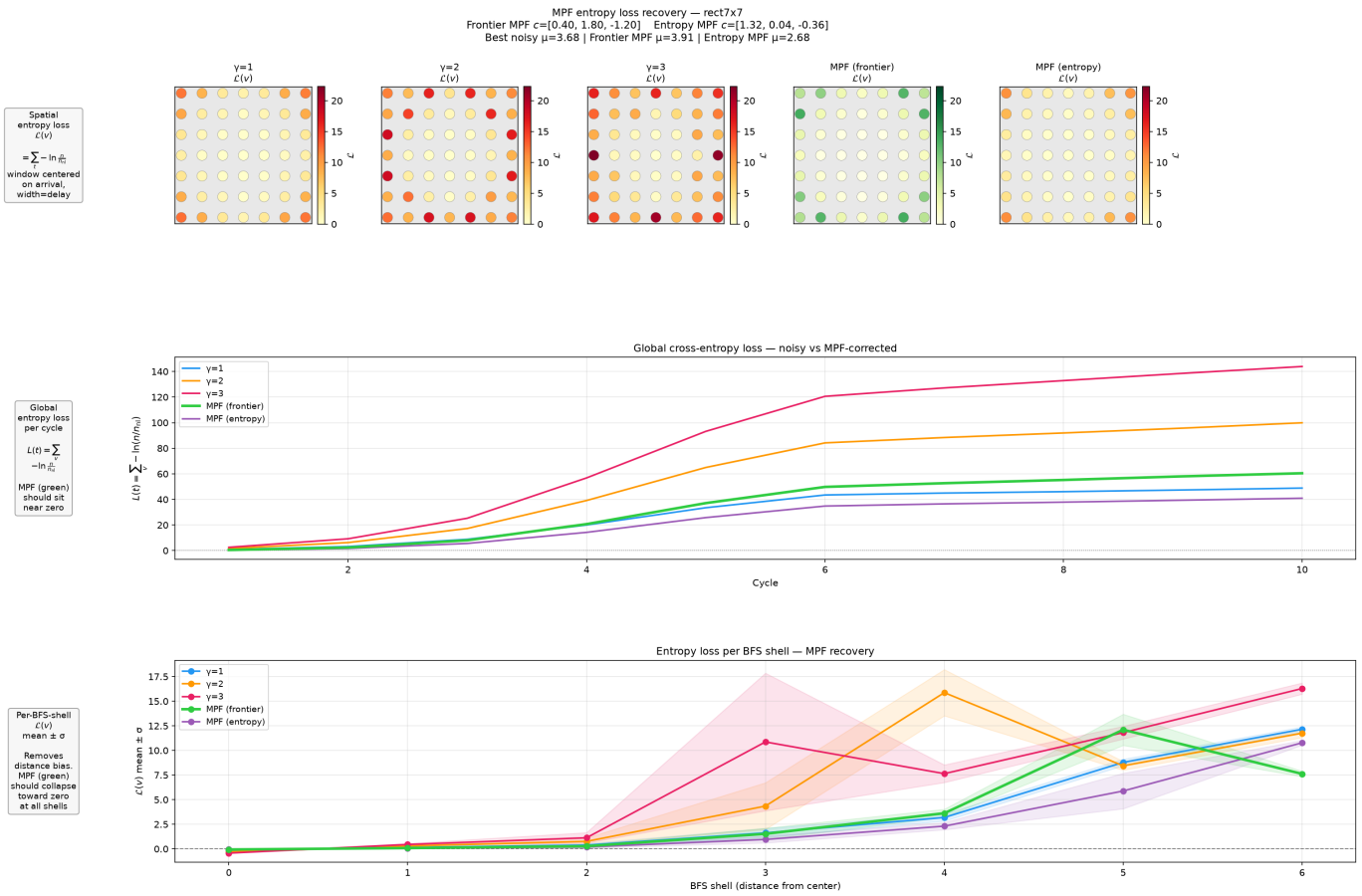


Figure 5: Entropy loss wavemap — rect 7×7, uniform depolarizing noise. Row 1: spatial map of $\mathcal{L}(v)$ per site for $\gamma = 1, 2, 3$, frontier MPF, and entropy-optimal MPF (purple). Row 2: global cross-entropy loss $L(t)$ per cycle. Row 3: mean $\pm\sigma$ of $\mathcal{L}(v)$ per BFS shell — cleanly separated γ curves with monotone ordering; entropy MPF sits below all noisy curves.

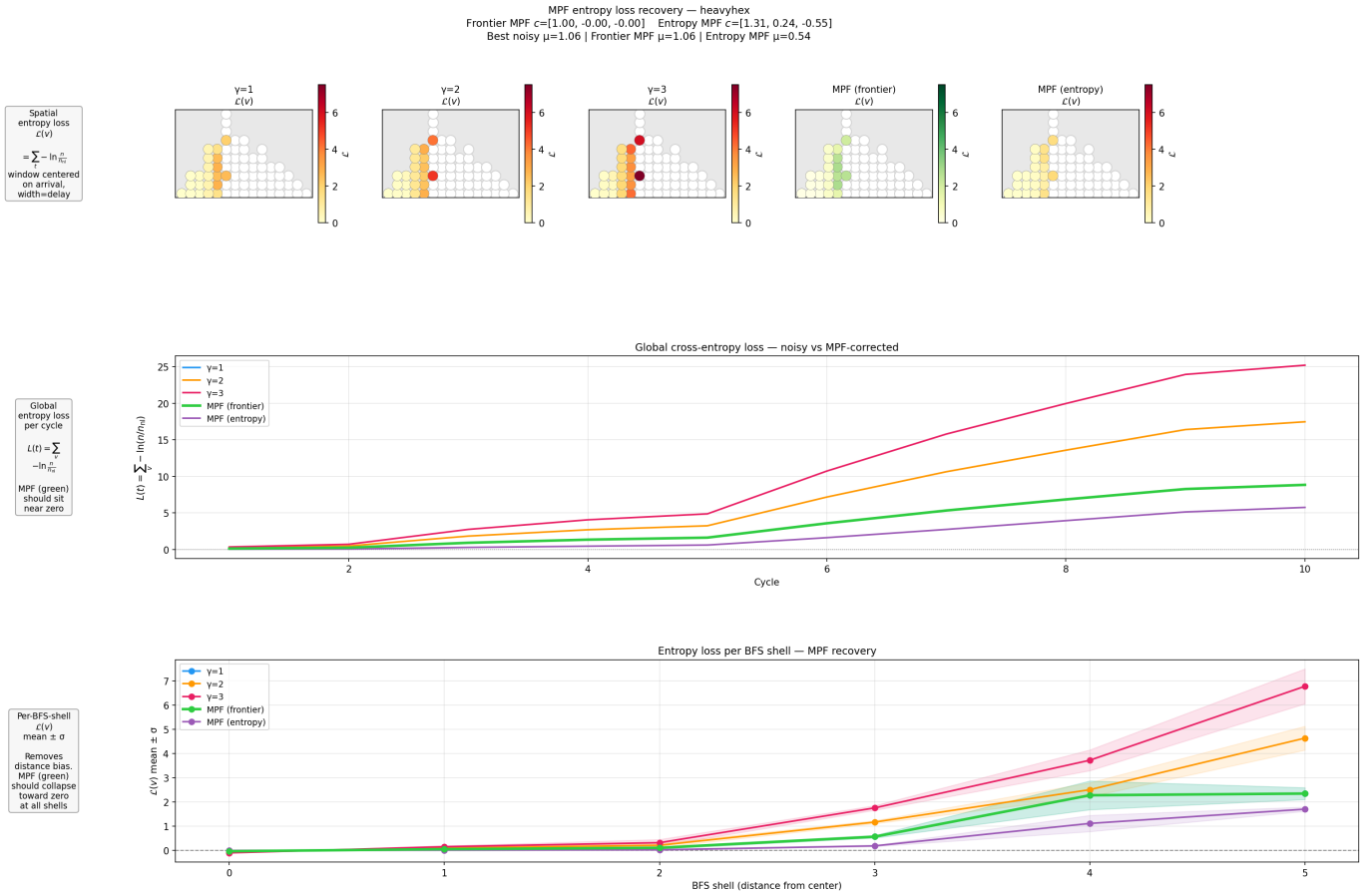


Figure 6: Entropy loss wavemap — heavy-hex 3×3, uniform depolarizing noise (control experiment). Same layout as the rect 7×7 figure. Despite only 21/68 sites inside the noiseless lightcone, the uniform noise produces cleanly separated per-shell entropy loss curves (monotone ordering) and 49% entropy recovery under the LR constraint ($\mu_{\mathcal{L}} : 1.058 \to 0.538$). Comparing with the hardware noise result (Figure 7) isolates the cause: per-bond PTM heterogeneity, not sparse lightcone coverage, is responsible for the information-starved regime.

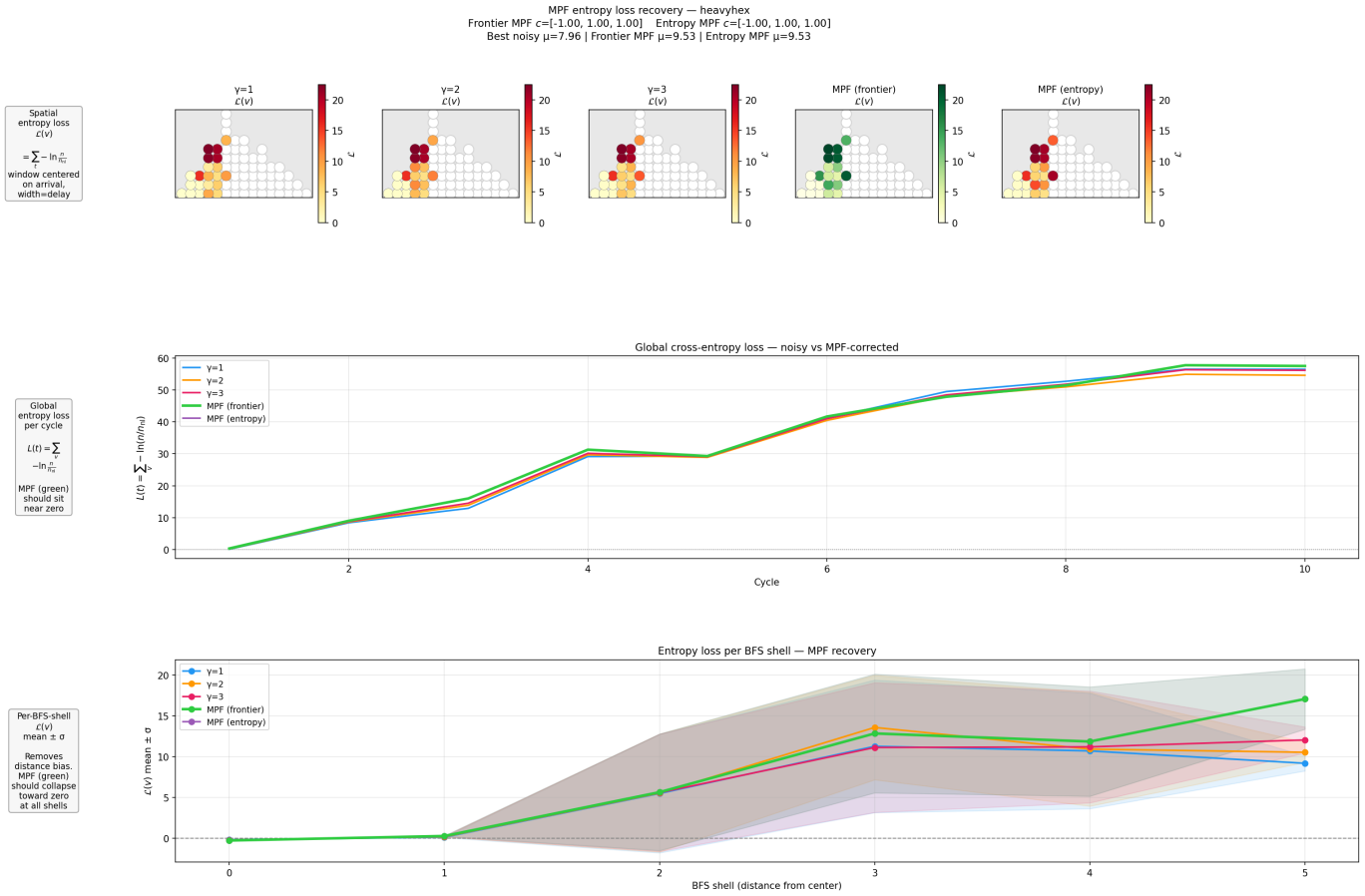


Figure 7: Entropy loss wavemap — IBM heavy-hex 3×3 lattice ($\chi = 70$, cutoff $\delta = 10^{-3}$, 10 cycles, center site 13). Same four-row layout as the rect 7×7 figure. Only 21/68 sites are inside the noiseless lightcone at 10 cycles — the wavemap is sparse. The per-shell entropy loss (Row 4) shows nearly identical curves for all three γ values, consistent with the information-starved regime where the LR constraint is already binding.

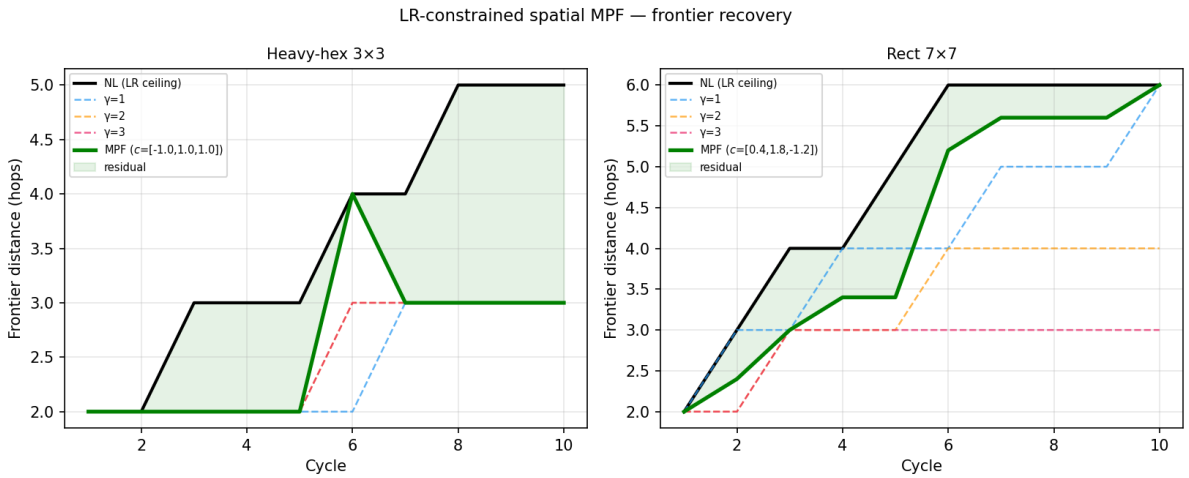


Figure 8: LR-constrained spatial MPF frontier recovery. Black: noiseless $F_{\mathrm{nl}}(t)$ (Lieb-Robinson ceiling). Colored: noisy frontiers $F_{\gamma_i}(t)$. Dashed: MPF-corrected $F_{\mathrm{lr}}(t) = \sum_i c_i F_{\gamma_i}(t)$ subject to $0 \le F_{\mathrm{lr}}(t) \le F_{\mathrm{nl}}(t)$. Left: rect 7×7 ($c \approx [+2, -1, 0]$) — clean recovery within 1 hop. Right: heavy-hex ($c \approx [-2, +1, +2]$) — negative c_1 encodes the inverted deficit ordering.

survivorship bias and traces how information loss accumulates with propagation distance, independently of which sites the noisy wave manages to reach.

Noise amplification γ as an effective Trotter rescaling. The eigenvalue analysis (Table 1) shows that the noise channel M_γ rescales the eigenvalues of the composed PTM without changing its eigenvectors:

$$\mathrm{PTM}_{\mathrm{noisy}}(\gamma) \approx V \cdot \big(\Lambda - \delta\Lambda(\gamma)\big) \cdot V^{-1} \tag{10}$$

where V are the gate eigenvectors (independent of γ) and $\delta\Lambda(\gamma)$ is a small diagonal correction scaling with γ. This is structurally identical to reducing the Trotter step: the ideal gate eigenvalues also scale with ε as $\lambda_i(\varepsilon) \approx \lambda_i(0) \cdot g(\varepsilon)$. Noise acts as a slower effective Trotter step: increasing γ reduces $\varepsilon_{\mathrm{eff}}$, so the wave propagates more slowly. Conversely, noise amplification ($\gamma > 1$) corresponds to an even smaller $\varepsilon_{\mathrm{eff}}$, making the wave slower still — the opposite direction from varying ε directly in ε-MPF, where a larger Trotter step makes the wave faster.

$$\mathrm{PTM}_{\mathrm{noisy}}(\gamma) \approx \mathrm{PTM}_{\mathrm{ideal}}(\varepsilon_{\mathrm{eff}}(\gamma)), \qquad \varepsilon_{\mathrm{eff}}(\gamma) < \varepsilon \tag{11}$$

Extrapolating $\gamma \to 0$ recovers $\varepsilon_{\mathrm{eff}} \to \varepsilon$, the ideal gate. Both γ-MPF and ε-MPF extrapolate back toward the ideal gate, but from opposite directions: ε-MPF varies the gate strength directly and monotonically, while γ-MPF varies it indirectly through the composed noise-gate product, introducing bond-dependent corrections that reduce conditioning.

The wavemap delay $l_\gamma(v)$ is a spatial, site-resolved measurement of this effective rescaling: it records how many extra cycles the noisy wave needs to reach threshold at each site, relative to the ideal gate. The site-by-site variation of $l_\gamma(v)$ (Table 3) reflects the bond-dependent corrections $\delta\Lambda(\gamma)$ — different bonds contribute different eigenvalue shifts depending on their hardware PTM, making the effective slowdown spatially non-uniform even for nominally uniform noise amplification.

Connection to MPF and noise learning. The Lindblad MPF framework [8] learns per-bond noise rates from hardware measurements and uses them to construct error-mitigated observables. Our wavemap

is complementary: rather than learning a noise model, we directly measure its spatial effect on operator propagation. The per-site ratio $r_\gamma(v)$ is in principle invertible to per-bond damping rates along dominant propagation paths — a model-free tomography from wavefront measurements alone.

Gate structure vs noise structure. A key finding is that the Pauli mixing structure of the wave (off-diagonal PTM fraction $\approx 36\%$) is entirely from the gate and independent of noise level. This means the spatial pattern of operator spreading is a property of the *Hamiltonian*, not the noise. Noise acts as a transparent filter that delays and damps the gate-determined wave without redirecting it. This clean separation validates the wavemap as a noise diagnostic: the spatial delay and ratio fields encode noise effects on top of a fixed gate-determined propagation template.

Future Work

Several directions remain open. The wavemap observables $l_\gamma(v)$ and $\mathcal{L}_\gamma(v)$ depend on the choice of center site; a systematic study across multiple centers would establish whether the per-shell distributions are universal properties of the topology and noise model, or depend on the initialization point. Longer simulations (20+ cycles) would reveal how the wavemap evolves after the wavefront reflects from the lattice boundary. On the simulation side, the arrival time $t_{\mathrm{nl}}(v)$ provides a per-site lightcone-aware memory allocation schedule that could reduce tensor network memory by 3–13×; and the frontier velocity v_{hop} predicts the onset of belief propagation approximation error via $t_{\mathrm{loop}} = g/(2v_{\mathrm{hop}})$, without requiring an expensive chi sweep.

The most promising longer-term direction is noise classification via machine learning. CppSim can generate a labeled training dataset at scale: sweep over a library of noise models (pure damping, rotation, heterogeneous per-bond, correlated errors), compute the wavemap for each, and label each simulation with the noise type and per-bond parameters. A classifier trained on this dataset learns the map from wavemap signatures to noise types; a graph neural network operating on the lattice graph is the natural architecture, as the wavemap is itself a spatial field over the graph. Deployed on a quantum processor: measure $n(v,t)$ via randomized benchmarking, compute the wavemap, run the classifier — noise type and severity identified without further simulation or analytical inversion. CppSim in this role is a noise characterization laboratory, generating the ground truth that no experiment alone can provide. The wavemap observables defined here are the features; the frontier paper is the foundation.

Finally, $n(v,t)$ is measurable on a quantum processor via randomized benchmarking or Pauli noise tomography, which would allow direct experimental validation of the wavemap on hardware and close the loop between classical simulation and quantum experiment.

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